Module 4 challenge

1st Attempt:

1. A data professional wants to merge two pandas dataframes. They want to join the data so only the keys that are in both dataframes get included in the merge. What technique can they use to do so?

1 / 1 point

- (selected) Inner join
- Left join
- Right join
- Outer join

Nice work

2. A data professional is working with two Python sets: set A and set B. What function can they use to find only the elements present in set A that are not also in set B?

1 / 1 point

- `union()`
- `intersection()`
- `symmetric_difference()`
- (selected) `difference()`

Nice work

3. **In pandas, what is the difference between the `iloc[]` and `loc[]` methods?**

1 / 1 point

- `iloc[]` merges two dataframes vertically; `loc[]` merges two dataframes horizontally.
- `iloc[]` selects dataframe rows and columns by name; `loc[]` selects dataframe rows and columns by index.
- `iloc[]` selects dataframe rows and columns by index; `loc[]` selects dataframe rows and columns by name.
- `iloc[]` merges two dataframes horizontally; `loc[]` merges two dataframes vertically.

Nice work

4. **Which of the following statements accurately describe Python lists? Select all that apply.**

1 / 1 point

- [ ] Lists are immutable.
- [x] Lists can contain sequences of elements of any data type.

Nice work

- [x] Lists can be indexed and sliced.

Nice work

- [x] Lists are mutable.

Nice work

5. **A data professional is working with a pandas dataframe named `sales` that contains sales data for a retail website. They want to know the price of the most expensive item. What code can they use to calculate the maximum value of the `Price` column?**

1 / 1 point

- `sales['Price'].max()`
- `sales.max().Price`
- `sales.max().[Price]`
- `sales = 'Price'.max()`

Nice work

6. **In Python, which of the following characters can a data professional use to instantiate a dictionary?** 1 / 1 point

- `< >`
- `{ }`
- `( )`
- `[ ]`

Nice work

7. **Where are modules accessed in Python?** 1 / 1 point

- Within a set
- Within a package or library
- Within a dictionary
- Within a global variable

Nice work

8. **A data professional is working with a dictionary named `employees` that contains employee data for a healthcare company. What Python code can they use to retrieve both the dictionary's keys and values?** 1 / 1 point

- `employees.keys()`
- `keys.employees()`
- `employees.items()`
- `items.employees()`

Nice work

9. **Which of the following statements accurately describe Python tuples? Select all that apply.** 1 / 1 point

- [x] Tuples are immutable.

Nice work

- [x] Tuples can be split into separate variables.

Nice work

- [ ] Tuples cannot be split into separate variables.
- [x] Tuples are sequences.

Nice work

10. A data professional is working with a list named `cities` that contains data on global cities. What Python code can they use to add the string `'Tokyo'` to the end of the list?

1 / 1 point

- `cities.pop('Tokyo')`
- `cities.append('Tokyo')`
- `cities.import('Tokyo')`
- `cities.insert('Tokyo')`

Nice work

11. Fill in the blank: A _____ NumPy array can be created from a list of lists, where each internal list is the same length.

1 / 1 point

- two-dimensional
- four-dimensional
- one-dimensional
- three-dimensional

Nice work